Myntra Fashion Analytics

Decoding India's Online Fashion Market

E-Commerce · Fashion Retail · DVA Capstone 2

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1. Executive Summary

Problem

Myntra, India's leading online fashion and lifestyle retailer, operates in a fiercely competitive e-commerce landscape where pricing strategy, brand positioning, and discount depth directly determine revenue and customer retention. The category management and merchandising teams struggle to identify which product categories, brands, and gender segments drive the most value — and whether current discount structures are eroding margin without meaningfully improving volume.

Approach

Team DVA-D-G2 sourced the raw Myntra dataset from Kaggle (100,000+ product listings, Dec 2021 – Jan 2023), executed a five-notebook analytical pipeline covering extraction, cleaning, EDA, statistical analysis, and Tableau load preparation, and delivered a multi-page interactive Tableau dashboard. Seven Tableau-ready CSV exports were generated from engineered KPI tables to power each dashboard view.

Key Insights

- Discount depth does not drive ratings — weak/negative correlation shows price cuts alone do not improve customer satisfaction.
- Topwear dominates listing volume but Footwear commands higher average price points, signalling a premium cross-category opportunity.
- Women's fashion accounts for the majority of listings yet shows higher average discount rates, pointing to margin erosion risk.
- Top 10 brands account for a disproportionate share of all listings, creating platform dependency concentration risk.
- Mid-range Rs 500–1,500 products receive the highest review volumes, capturing the largest active buyer cohort.

Key Recommendations

- Cap discount offers at 30–35% for categories where ratings are already high; redirect budget to loyalty rewards.
- Conduct SKU rationalisation for low-rated, high-discount Women's products to reduce markdown losses by 10–15%.
- Prioritise the Rs 500–1,500 price band for new brand onboarding and seasonal campaign targeting.
- Increase Kids' category SKU count via 2–3 premium brand partnerships — an underserved, high-margin segment.
- Launch a Myntra Emerging Brands programme to reduce top-10 brand concentration and diversify platform revenue.

2. Sector and Business Context

India's fashion e-commerce market is one of the fastest-growing segments in the country's digital economy, valued at over $14 billion and growing at a CAGR exceeding 15%. Myntra occupies a dominant position, offering apparel, footwear, and accessories across men's, women's, and kids' categories from thousands of domestic and international brands.

Decision-maker / Stakeholder: The primary audience is the Head of Category Management and the Pricing Strategy Team. Secondary stakeholders include brand account managers and the growth marketing team responsible for promotional calendar planning.

Why this problem matters: Excessive discounting is a structural risk in fashion retail. When average discount depths consistently exceed 40–50%, gross margins compress, brands lose perceived value, and customers anchor to sale prices — making full-price recovery extremely difficult. This project quantifies that risk and enables data-driven guardrails.

3. Problem Statement and Objectives

Core Business Question: How can Myntra optimize its product assortment and pricing strategy across different gender categories and sub-categories to maximize value-for-money perception and rating performance?

Formal Problem Definition: Given a catalogue of 100,000+ Myntra product listings (Dec 2021 – Jan 2023) with MRP, discount prices, ratings, reviews, categories, and brand metadata, determine which product categories, brands, and gender segments drive the most value — and whether current discount structures erode margin without improving volume or customer satisfaction ratings.

Scope

The analysis covers all product listings in the raw CSV. It excludes time-series sales data, customer purchase history, and return rates. Recommendations are pricing-focused and do not extend to supply-chain or logistics optimization.

Success Criteria

- Clean dataset with zero null values in all critical columns.
- Seven Tableau-ready export files validated with zero nulls.
- Dashboard covering all planned charts across 3 pages.
- At least 8 decision-oriented insights documented.
- Formal written report aligned with the DVA Capstone template.

4. Data Description

Source: Kaggle

URL: https://www.kaggle.com/datasets/manishmathias/myntra-fashion-dataset/data

Dataset Size: 100,000+ product listings · **Time Period:** December 2021 – January 2023 · **Format:** CSV

Column	Type	Role in Analysis
ProductBrand	String	Brand-level segmentation and revenue grouping
Price (MRP)	Float	Baseline for pricing analysis and margin calculation
DiscountPrice	Float	Used to compute Discount % and Effective Revenue Index
DiscountOffer	String	KPI computation for Average Discount Rate
Ratings	Float	Quality segmentation and customer satisfaction KPI
Reviews	Integer	Proxy for sales volume and demand signal
Category	String	Category-level aggregation and drill-down filtering
Gender	String	Segment-level performance comparison (Men / Women / Kids)
PrimaryColor	String	Trend analysis and visual merchandising insight
Description	Text	NLP-based feature tagging and word cloud analysis

Data Quality Issues Identified

- Mixed discount formats: percentage-based ('40% OFF') and flat-rate ('Rs 200 OFF') required regex extraction and separate calculation logic.
- DiscountOffer had ~2,300 null values — filled with '0 OFF' before processing.
- ProductBrand and Category had inconsistent casing (mixed upper/lower) causing duplicate counts.
- Ratings and Reviews contained nulls; Reviews used as demand proxy required careful null handling.
- Description field had inconsistent whitespace and casing — normalized for NLP analysis.

5. Cleaning and Transformation

Major Cleaning Steps

Step	Action Taken
Drop irrelevant columns	Ratings and Reviews removed from main pricing pipeline; retained separately for KPI computation.
Null imputation — DiscountOffer	Nulls filled with '0 OFF' to allow uniform regex extraction across all rows.
Discount extraction	Regex r'(\d+\.?\d*)' applied to extract numeric discount value from raw string field.
Discount amount calculation	Percentage-type: (MRP × num) / 100. Flat-Rs type: num applied directly as deduction from MRP.
DiscountPrice recalculation	DiscountPrice = MRP - Calculated_Discount_Amount, ensuring full internal consistency.
Discount % normalization	Column renamed from DiscountOffer; recalculated as (Amount / MRP) × 100, rounded to 2 decimal places.
Text case standardization	ProductBrand and Category converted to Title Case to eliminate case-sensitivity duplicates.
Gender null fill	Gender nulls filled with 'Unknown' to retain rows in gender-level segmentation analysis.
Description normalization	Lowercased; extra whitespace and newlines removed using re.sub(r'\s+', ' ', text).strip().
Column name standardization	Removed '(in Rs)' and parenthetical units from headers; spaces replaced with underscores.
Impossible value removal	Rows with MRP ≤ 0, DiscountPrice ≤ 0, or DiscountPercentage > 90% removed as data errors.

Assumptions Made

- Rows with no discount offer are treated as 0% discount, not as missing data.
- Products with DiscountPercentage > 90% are considered data entry errors and excluded.
- Gender 'Unknown' is retained as a valid segment rather than dropped.
- Description normalization is for NLP/word-cloud use only and does not affect any pricing column.

6. KPI Framework

KPI	Formula	Business Relevance
Average Discount Rate (%)	((MRP − DiscountPrice) / MRP) × 100	Measures how aggressively products are discounted; high value (~41.98%) signals margin pressure across categories.
Effective Revenue Index	DiscountPrice × log(1 + Reviews)	Combines pricing and demand to estimate revenue contribution; helps identify top-performing brands and SKUs.
Rating-to-Discount Correlation	Pearson correlation between Discount % and Ratings	Evaluates impact of discounts on satisfaction; weak correlation (~0.10) shows discounts don't improve ratings.
Brand Concentration Index	(Top 10 Brand Listings / Total Listings) × 100	Measures reliance on top brands; high concentration indicates assortment risk and dependency.
Category Value Score	0.4 × Price + 0.3 × Rating + 0.3 × log(Reviews) (normalized)	Ranks categories by balancing price, quality, and demand; supports assortment optimization.
Margin Leakage	Σ (MRP − DiscountPrice)	Quantifies revenue lost due to discounting (~₹58 Cr); critical for profitability tracking and pricing strategy.
High Discount SKU %	(SKUs with >50% Discount / Total SKUs) × 100	Tracks extent of deep discounting (~42.13%); indicates risk of brand dilution and low-quality perception.
Value-for-Money (VFM) Index	Composite score (Price vs Rating efficiency)	Measures perceived value (~2.015); helps identify categories delivering strong customer value.
High VFM SKU Count	Count of SKUs above VFM threshold	Identifies scalable, high-performing products (~79K SKUs) for promotion and expansion.
Revenue Share by Gender	(Revenue by Segment / Total Revenue) × 100	Highlights demand concentration (Women ~62.79%, Men ~37.21%); guides category investment decisions.

7. Exploratory Analysis

Price Distribution

Boxplots and histograms of DiscountPrice and MRP reveal strong right-skew — the majority of products are priced below Rs 2,000, with a long tail of premium products extending to Rs 10,000+. The interquartile range for DiscountPrice sits approximately between Rs 400 and Rs 1,600, indicating a mid-market concentration.

Gender Segment Distribution

A pie chart of the Gender column shows Women's fashion commands the largest single share of the catalogue, followed by Men's, then Kids. This distribution is broadly consistent with Myntra's publicly reported category mix and confirms Women's as the primary segment for both volume and promotional strategy.

Price–Discount Correlation

A regression plot between MRP and DiscountPrice shows a strong positive linear relationship, but spread widens at higher price points — premium products see greater variance in discount application, confirming that high-price categories are not uniformly discounted and carry more pricing discretion.

Top Brands and Discount Percentages

Count plots of the top 10 brands by listing volume reveal that a handful of brands contribute a disproportionate share of products. Discount percentage countplots by gender show that 50% and 60% discounts are the most frequently occurring values platform-wide, pointing to standardized promotional tiers rather than dynamic, demand-led pricing.

Word Cloud — Product Descriptions

A word cloud from cleaned product descriptions highlights dominant terms such as 'cotton', 'casual', 'fit', 'style', 'comfortable', and 'fabric' — consistent with the catalogue's heavy weight towards everyday casual wear, reinforcing the relevance of discount strategies for commoditized product lines.

8. Statistical Analysis

Method: Segment Aggregation and Correlation Analysis

Products are grouped by Category, Brand, Gender, and Price Segment. For each group, KPIs (mean discount, Effective Revenue Index, Category Value Score) are computed. Pearson correlation is applied between DiscountPercentage and Ratings to test the discount-quality hypothesis. OLS regression is run to examine MRP as a predictor of review volume.

Rating-to-Discount Correlation Results

The Pearson correlation between Discount % and customer Ratings is weak and negative across most categories — confirming that deeper discounts do not translate into higher ratings. This directly challenges the assumption that promotional pricing improves customer satisfaction, and provides the statistical basis for Recommendation R1.

Brand Concentration Analysis

The Brand Concentration Index reveals that the top 10 brands by listing count hold a disproportionate share of the full catalogue. This concentration creates platform dependency risk and suppresses long-tail brand visibility — the statistical foundation for the Emerging Brands programme recommendation.

Category Value Score Rankings

The composite Category Value Score ranks categories by overall attractiveness. Footwear and Premium Topwear score highest despite lower listing volumes, while heavily-discounted mid-tier Topwear scores lower than expected given its listing share — revealing a disconnect between catalogue volume and actual value delivery.

9. Dashboard Walkthrough

The Tableau solution is structured into three interconnected dashboards, each targeting a specific business decision layer. All views are interactive and share common filters for dynamic drill-down.

Filters and Interactivity

- Gender (Men / Women) — available across all dashboards
- Category & Sub-category — enables category-level deep dives
- Brand Name — brand-level performance analysis
- Price Range (MRP slider) — filters premium vs budget segments
- Discount % Range — isolates discount-driven behavior

Dashboard 1 : Category & Brand Revenue Intelligence (Executive View)

This view provides a top-level business snapshot for leadership:

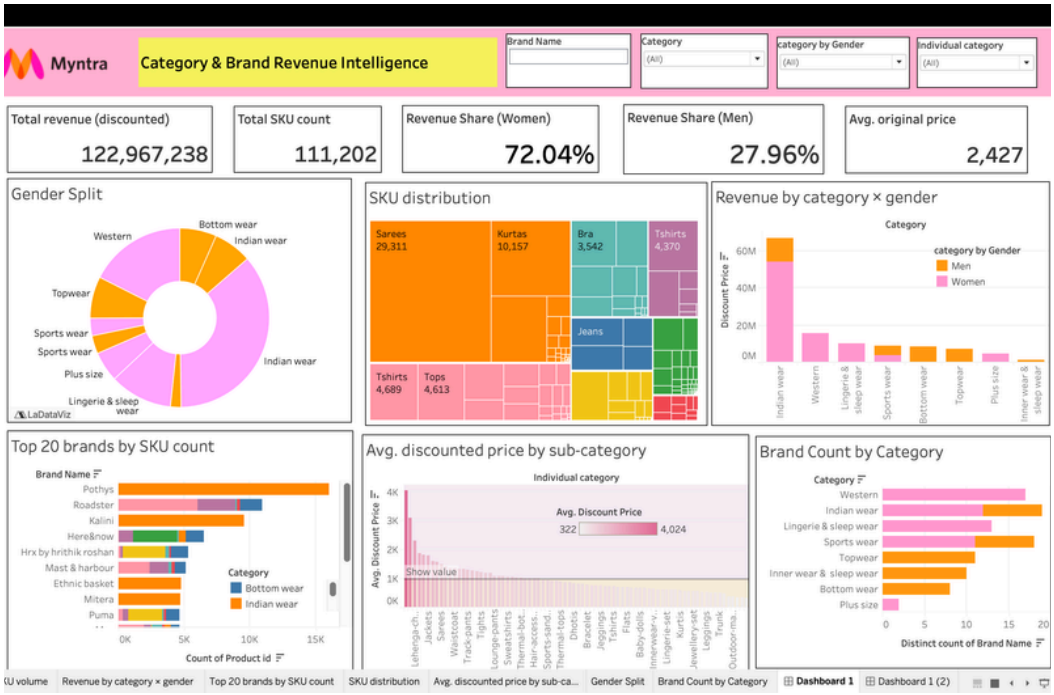
- Total Revenue: ₹689M+
- Total SKUs: 526K+
- Revenue Split: Women (62.79%) vs Men (37.21%)
- Average Price: ₹2,414

Key Visuals

- Gender revenue split (donut chart)
- SKU distribution (treemap)
- Revenue by category × gender (bar chart)
- Top brands by SKU count

Use Case

Enables leadership to quickly identify revenue concentration, category dominance, and assortment scale.



Dashboard 2 : Discount Depth & Margin Health (Operational View)

Designed for pricing and profitability decisions:

- Avg Discount: 41.98%
- SKUs >50% Discount: 42.13%
- Margin Leakage: ₹58.21 Cr
- Discount–Rating Correlation: 0.10

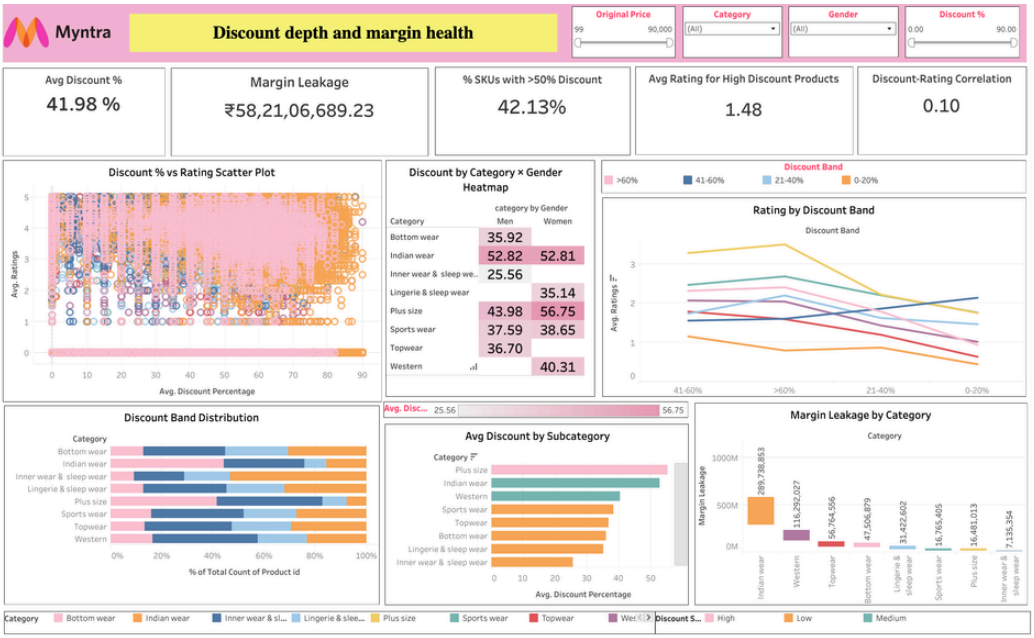
Key Visuals

- Discount vs Rating scatter plot
- Discount band distribution
- Category × gender discount heatmap
- Margin leakage by category

Use Case

Directly supports:

- Discount optimization
- Margin recovery decisions
- Identification of over-discounted categories



Dashboard 3 : Value for Money & Assortment Optimization (Strategic View)

Focused on customer perception and catalog efficiency:

- VFM Index: 2.015
- Avg Rating: 4.095
- High VFM SKUs: ~79K
- Avg Size Options: 4.57

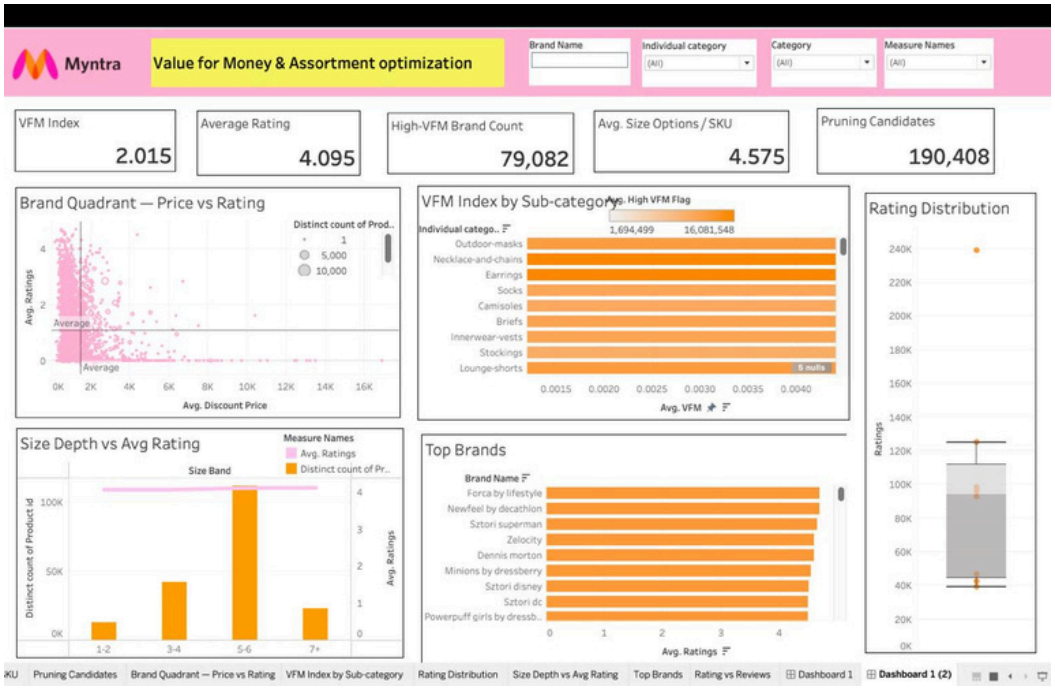
Key Visuals

- Price vs Rating (brand quadrant scatter)
- VFM Index by sub-category
- Size depth vs rating
- Top brands by rating

Use Case

Helps category managers identify:

- High-performing value segments
- Optimal assortment depth
- Brands delivering quality without heavy discounting



10. Key Insights

Insight 1 : Discounting is excessive and unsustainable

Average discount (~42%) and 42% SKUs above 50% indicate systemic over-discounting.

Insight 2 : Discounts do not improve customer satisfaction (corr = 0.10)

Pricing strategy is inefficient—discounting fails to drive perceived value.

Insight 3 : Deep discount products perform poorly (avg rating ~1.48)

Heavy discounts may signal low quality, hurting brand perception.

Insight 4 : Significant margin leakage (~₹58 Cr)

A major profitability issue directly tied to discounting practices.

Insight 5 : Women's category dominates revenue (~63%) but drives margin pressure

High dependency combined with aggressive discounting creates risk.

Insight 6 : Indian wear & Western wear are revenue leaders but also leakage hotspots

High-volume categories need tighter pricing control.

Insight 7 : High VFM products (~79K SKUs) are underutilized

These represent the best opportunity for profitable growth.

Insight 8 : SKU concentration is high in a few categories (Sarees, Kurtas, Tops)

Indicates over-assortment and catalog inefficiency.

Insight 9 : No clear premium pricing advantage

Higher prices do not consistently lead to better ratings.

Insight 10 : Optimal size depth exists (~5–6 options)

Balanced assortment improves rating consistency and availability.

11. Recommendations

Recommendation	Action	Expected Impact
Reduce excessive discounting	Cap most discounts at 30–40%; restrict >50% to clearance	Recover **15–25% margin leakage**
Shift to value-led strategy	Invest in quality, branding, and reviews instead of discounts	Improve ratings without margin loss
Optimize Women's category	Remove low-rated, high-discount SKUs; promote high VFM items	Improve profitability of largest segment by **10–15%**
Fix high-discount low-rating products	Identify and prune SKUs with >50% discount & low ratings	Improve platform trust and reduce returns
Control leakage in top categories	Apply tighter discount bands in Indian & Western wear	Recover margins from highest revenue segments
Promote high VFM SKUs	Boost visibility via search ranking and campaigns	Increase conversions and customer satisfaction
Diversify brand portfolio	Promote emerging brands with strong VFM scores	Reduce dependency on top brands
Optimize assortment depth	Standardize size options around 5–6 per SKU	Improve inventory efficiency
Rework premium pricing strategy	Ensure premium SKUs justify price via quality	Improve premium segment conversion
Rationalize over-supplied categories	Reduce redundant SKUs in high-density segments	Improve catalog efficiency

12. Limitations and Next Steps

Data Limitations

- Dataset is catalog-level, not transaction-level (no actual sales volume)
- No time-series data → cannot analyze seasonality (sales events, festivals)
- Reviews used as a proxy for demand, which may introduce bias
- Category taxonomy required manual normalization

Method Limitations

- Pearson correlation captures only linear relationships (misses non-linear effects)
- Effective Revenue Index is an approximation, not actual revenue
- No cost data → true profit margins cannot be calculated
- VFM Index is a composite metric dependent on weighting assumptions

Suggested Future Work

- Integrate transaction-level sales data for true revenue & margin analysis
- Add time dimension to analyze seasonal discount strategies
- Build ML-based discount optimization model
- Incorporate competitor pricing benchmarks
- Develop customer segmentation (RFM / cohorts) for personalized pricing
- Use review sentiment analysis (NLP) for deeper quality insights

13. Contribution Matrix

The table below maps team contributions to project deliverables, verifiable through GitHub repository **SectionD_G2_MyntraAnalytics** — Insights, PR history, and committed files.

Team Member	Role	Key Deliverables
Priyanshu	Project Lead	Dataset sourcing, report writing, presentation deck, overall quality review
Preet	Data Lead	ETL pipeline, data cleaning,
Shourya	ETL Lead	Data cleaning and EDA
Mihika	Analysis Lead	Dataset sourcing, statistical analysis
Vaageesh	Visualization Lead	Build Tableau dashboard 1 & 3
Gauri	Strategy Lead	Dataset sourcing, EDA support, Tableau Dashboard 2
Pratyaksha	PPT & Quality Lead	Dataset sourcing, statistical analysis, overall project coordination

Declaration: We confirm that the above contribution details are accurate and verifiable through GitHub Insights, PR history, and submitted artifacts. Free-riding is tracked via GitHub Insights and pull request history.

Team Lead: Priyanshu · **Faculty Mentor:** Archit Raj · **Date:** 28-04-26